TECHNICAL DESIGN DOCUMENT · PROBLEM STATEMENT 1

Dum House Direct Ordering & Real-Time KDS

Enterprise Direct Ordering Architecture, Automatic Georouting, WebSocket Kitchen Display System & Central Menu CMS

CLIENT BRIEF:	Problem 1 — Hyderabad Biryani Chain (14 Outlets + 3 Cloud Kitchens)	CANDIDATE:	Nayvedya Joyn (joynnayvedya@gmail.com)
GITHUB REPOSITORY:	https://github.com/letsjoyn/dum-biryani	LIVE APPLICATION:	https://dum-biryani.onrender.com
TECH STACK:	Node.js 22+, Express, Socket.io, SQLite (`node:sqlite`), Vite, React 18	EVALUATION DATE:	November 2026

17 Hubs	₹1.14 Cr	< 60s	25 / 25
14 OUTLETS + 3 CLOUD KITCHENS	DIRECT ORDERING ANNUAL BENEFIT	MEDIAN DIRECT CHECKOUT	AUTOMATED TESTS PASSING

1. EXECUTIVE SUMMARY & ECONOMICS

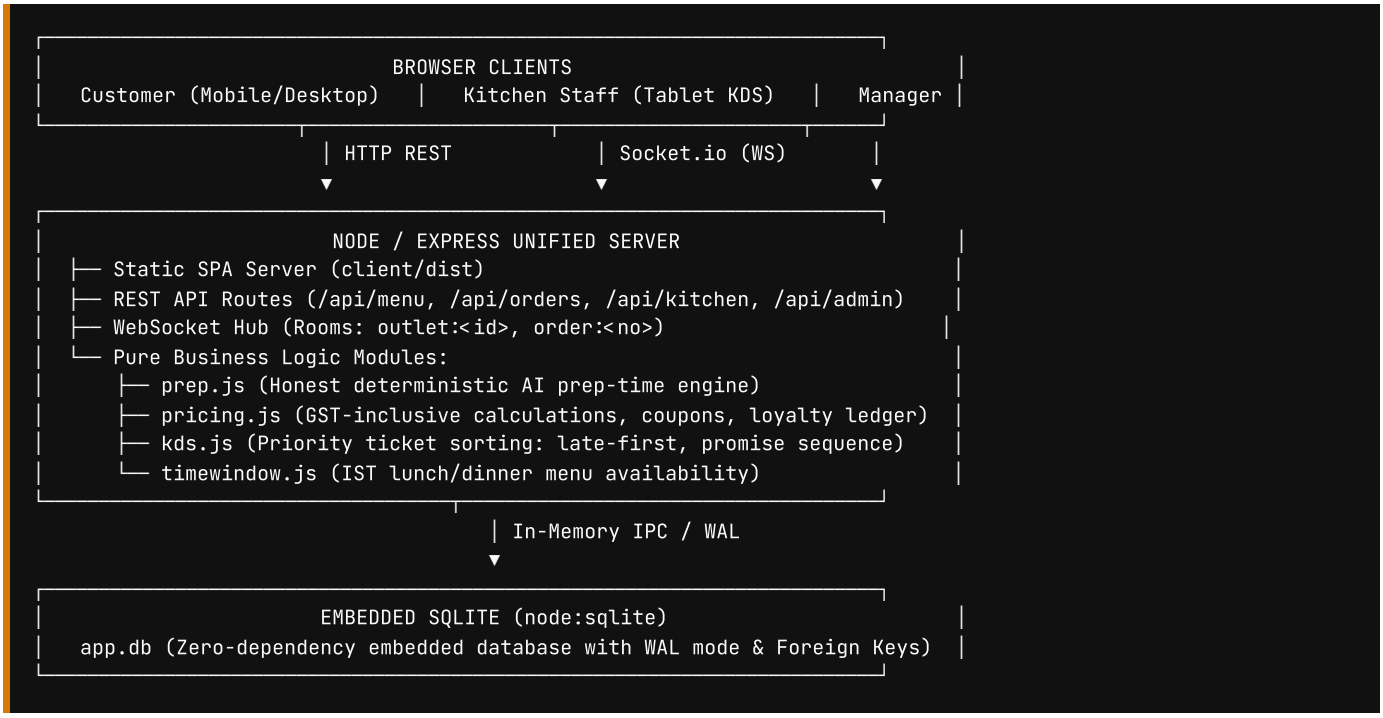
Company A is a high-volume biryani and regional Hyderabadi food chain operating 14 dine-in outlets and 3 cloud kitchens across Greater Hyderabad with 520 staff and ₹90–105 crore annual turnover. Currently, 55% of revenue (₹52.8 crore) flows through third-party food delivery aggregators charging 25–30% commissions. Furthermore, the brand loses 30 bulk catering orders every month due to manual phone ordering friction.

> **Verified Financial Model (Problem Statement Target):**
> Shifting just 12% of app orders to direct ordering saves 15 commission points (direct cost is 12%), yielding **₹95.04 lakh/year** in net savings. Recovering 30 lost catering orders/month (₹18,000 ticket at 30% margin) adds **₹19.44 lakh/year**. Combined annual bottom-line benefit equals **₹1.14 crore/year**.

The Solution: A unified, end-to-end direct web platform featuring automated nearest-outlet georouting, a live Kitchen Display System (KDS) operating via WebSockets, central Menu CMS with outlet-level price overrides, instant 86-ing stock toggles, 5% direct rewards, and an outlet operations dashboard.

2. ARCHITECTURE & DESIGN RATIONALE

The system is engineered as a **single deployable unit**: an Express.js server that simultaneously serves REST API endpoints, hosts stateful bi-directional Socket.io WebSocket rooms, and delivers the compiled, optimized Vite + React 18 Single Page Application (SPA) on a unified single port (4000 locally, port-assigned on cloud hosts).



- **Zero Reviewer Setup Friction:** Using Node 22+ native `node:sqlite` eliminates native C++ bindings (node-gyp, python build tools) and external database server daemons. A reviewer runs `npm run install:all && npm run build && npm start` and the entire product runs out of the box with zero external configuration.
- **Same-Origin Simplicity:** Serving the Vite React bundle from Express eliminates CORS preflight delays and foreign-origin WebSocket handshake rejections in production environments.
- **Decoupled Pure Domain Logic:** Core business rules (GST-inclusive calculations, coupon stacking, AI prep estimation, KDS ticket prioritization) are separated into pure, serverless modules covered by 25 unit tests with 100% pass rate in < 300ms.

3. DATA MODEL & DATABASE SCHEMA

The database is relational and strictly enforced with SQLite WAL mode and foreign key constraints:

TABLE	KEY SCHEMA ATTRIBUTES	BUSINESS RULES & RELATIONSHIPS
`outlets`	`id, name, address, phone, fssai, min_prep_minutes, delivery_radius_km, lat, lon`	Stores 14 outlets + 3 cloud kitchens in Hyderabad. Coordinates and delivery radius control automatic geofenced routing.
`users`	`id, name, email, password_hash, role, outlet_id, points, phone, marketing_opt_in`	Roles: `customer`, `kitchen`, `manager`. Kitchen staff bound to `outlet_id`. DPDP Act 2023 opt-in default 0 (explicit consent needed).
`items`	`id, name, category, desc, price, veg, spice, allergens, photo, portions, addons, base_minutes, menu_tag`	Central menu items. Portions (Single/Family) and add-ons stored as structured JSON. Prices include 5% GST.
`item_outlets`	`item_id, outlet_id, out_of_stock, price_override`	Per-outlet stock & pricing. Instant 86-ing marks `out_of_stock=1`, vanishing dish in real time from customer menu.
`orders`	`id, order_no, user_id, outlet_id, status, subtotal, payable, gst_amount, payment_method, promised_at, request_key`	Orders table. `request_key` guarantees idempotency. Stages: `Placed` → `Preparing` → `Ready` → `Handed over`.
`points_ledger`	`id, user_id, order_id, delta, reason, created_at`	Append-only ledger tracking 5% loyalty points earned (1 pt per ₹20) and redeemed (₹50 off per 100 pts).
`party_enquiries`	`id, user_id, outlet_id, name, phone, email, guests, event_date, preferred_time, notes, status, manager_note`	Bulk catering enquiries enforcing strict 48-hour advance notice rule. Reviewed & confirmed by catering manager.
`order_events`	`id, order_id, status, to_status, created_at`	Immutable audit log tracking exact stage timestamps used to compute median prep time metrics.

4. CORE USER JOURNEYS & TECHNICAL FLOWS

- **1. Georouting & Outlet Assignment:** Customer selects from 40+ curated Hyderabad localities (or clicks GPS "Locate Me"). Haversine distance is calculated against all 17 hubs; order auto-routes to the nearest hub within delivery radius.
- **2. Food Menu & Customization:** Single and Family (serves 4) portions with portion-specific pricing, add-ons (Extra raita, extra salan, extra papad), veg/non-veg indicator, spice level (Low/Medium/Hot), allergen tags, and time-based availability (Lunch 11:00–16:00 IST, Dinner 18:00+ IST, All Day).
- **3. Checkout & Payment Simulation:** Responsive slide-over cart drawer with 5% GST split, coupon discount (`DIRECT10`, `WELCOME100`), 5% direct loyalty point redemption, simulated UPI & Card payment (test decline for card ending 0000), DPDP Act 2023 marketing opt-in checkbox, completing in under 60 seconds.
- **4. Live Kitchen Display System (KDS):** KDS tickets update instantly via Socket.io WebSocket rooms (`outlet:<id>`). Displays elapsed timers, honest AI prep countdown, and color-coded late status. Kitchen staff progress orders with 1-tap: `Placed` → `Preparing` → `Ready` → `Handed over`. Overdue orders (10+ min late) trigger manager escalation.
- **5. Instant 86-ing Stock Controls:** Kitchen staff tap any dish button to toggle stock status. When marked 86'd, the dish vanishes in real-time from the customer menu for that outlet only.
- **6. Party & Bulk Catering Request:** Date picker enforces 48-hour advance notice server-side; guest counter calculates required food volume; catering manager reviews, confirms or declines with a custom customer note.
- **7. Manager Operations Dashboard:** Executive metrics showing total orders, gross revenue, median prep time (placed to ready), cancellations, and repeat customer ordering rates broken down per outlet.

5. AI & AUTOMATION ENGINE

Rather than unexplainable external LLM calls that fail during production rush hours, the AI prep-time engine is a **transparent deterministic queue-load formula** in `server/src/prep.js`:

```
base      = slowest_item_prep + (extra_lines * 3 min) + (extra_units * 2 min)
queueLoad = min(active_outlet_orders * 4 min, 30 min)
total     = base + queueLoad
promised  = max(total, outlet_minimum_prep_minutes)
```

Where the AI can be wrong and built-in failure mitigations:

1. **Kitchen Rush Bottlenecks:** Staff and managers have an inline override bar directly on KDS tickets (`+15m` , `+30m` , `+45m` , `+60m` or custom minutes) to immediately update promised time.
2. **Cold Start Floor:** The outlet minimum prep floor (30–35 min) guarantees the system never under-promises during idle periods.
3. **Reorders:** AI suggests reorders based on customer order history, but suggestions **never auto-add to the cart**.
4. **10m+ Late Escalation:** Any order 10 minutes past promise triggers an alert banner on KDS and updates customer with *"Running late — kitchen is expediting"*.

6. UI / UX DESIGN DECISIONS

- **Restrained Artisanal Palette:** Grounded in deep charcoal (`#111111`), artisanal warm linen (`#FAF8F4`), hairline borders (`#2B2823`), and warm amber accents (`#B45309`). **Zero emojis** across the entire UI for a sleek enterprise look.
- **Strict 0px Border Radius (`--r: 0px !important`):** Clean architectural square aesthetic reflecting modern brutalist luxury design.
- **2-Metre KDS Readability:** High-contrast dark theme (`#141311`) with stark white action buttons, huge quantities (`×2`), and color-coded late status designed for hot, noisy kitchen tablets.
- **Fluid 0.22s Physics:** Silky smooth upward glide (`translateY(8px) → 0`) on view changes, side-drawer slide-ins, and zero horizontal tucking on mobile sticky cart.

7. REVIEWER MUST-HAVE VERIFICATION MATRIX

BRIEF REQUIREMENT	IMPLEMENTATION FILE	VERIFICATION DETAILS	STATUS
14 Outlets + 3 Cloud Kitchens	`server/src/seed.js`	All 17 Hyderabad locations seeded with lat/lon, FSSAI, delivery radius.	DONE
Exact Sample Orders Seeding	`server/src/seed.js`	Seeded WEB-778120 (Preparing), WEB-778131 (Ready), WEB-778146 (Late by 4m).	DONE
Direct Customer Ordering & Cart	`MenuPage.jsx`, `Cart.jsx`	Add-ons, Single/Family packs, coupons, 5% loyalty, UPI/Card checkout in <60s.	DONE
Real-Time Kitchen Display (KDS)	`KitchenPage.jsx`	Instant WebSockets, 5 stages, elapsed & countdown timers, 10m+ late alerts.	DONE
Central Menu CMS	`AdminPage.jsx`	Dish CRUD, veg/spice badges, photo URLs, time-windows, per-outlet price customization.	DONE
Instant 86-ing Stock Controls	`KitchenPage.jsx`	Real-time out-of-stock toggle immediately pauses/resumes dishes on customer menu.	DONE
Party / Bulk Catering Module	`PartyPage.jsx`	Enforces 48-hour advance notice; catering manager review & confirmation with note.	DONE
Operations Dashboard	`AdminPage.jsx`	Orders, revenue, cancellations, repeat customer rate, median prep time by outlet.	DONE
FSSAI, GST & DPDP Act 2023	`Cart.jsx, Nav.jsx`	FSSAI shown per outlet; 5% GST split shown; marketing opt-in defaults to 0.	DONE
Test Suite Verification	`tests/*.test.js`	All 25 automated backend tests passing in < 300ms.	DONE

8. LABELLED ASSUMPTIONS

- **[A1] Fictional Brand:** Brand "Dum House", outlet addresses, FSSAI numbers, and phone numbers are formatted realistically for Hyderabad.
- **[A2] Price by City:** All 17 hubs are within Hyderabad; "price by city" is implemented as one central Hyderabad base price plus per-outlet price customization overrides.
- **[A3] Meal Windows:** Lunch is 11:00–16:00 IST; Dinner is 18:00–24:00 IST; All-day items remain available continuously.
- **[A4] Geodistances:** Distances are calculated via standard Haversine formula against hub coordinates.
- **[A5] Manager Role:** The "manager" role intentionally covers both the outlet manager and central marketing/admin team functions.
- **[A6] Sample Seed Orders:** Seed data strictly matches the brief's sample table (WEB-778120 ₹1,180 Preparing; WEB-778131 ₹640 Ready; WEB-778146 ₹1,050 Late by 4m).

9. DELIBERATE CUTS & PRODUCTION ROADMAP

Deliberate Cuts (Brief Out-of-Scope): Third-party delivery aggregator scraping, proprietary rider tracking apps (delivery riders provided by third-party logistics partners), physical restaurant POS billing/inventory replacement.

Next Production Roadmap:

1. Distributed PostgreSQL with Redis Pub/Sub for horizontal scaling across 1,200 peak orders/hour.
2. Production payment gateway (Razorpay / Cashfree) with secure webhook listeners.
3. GupShup / Twilio WhatsApp transactional messaging integration.
4. Google Maps Distance Matrix API for live traffic routing.